Elena Alegret Regalado

Zürich, Switzerland

elenaalegret@gmail.com elenaalegret.github.io in/elena-alegret elenaalegret Google Scholar

Profile

AI/ML researcher pursuing an **M.Sc. at ETH Zürich**, focusing on 3D computer vision and generative models. Conducted four research internships resulting in **two papers at 3DV 2026** (one as co-first author), developed in collaboration with researchers from **Google, Microsoft, ETH and TUM**.

Research Interests: 3D Generative Models, 3D Computer Vision, 3D Reconstruction and Understanding, Segmentation, Robotics, Autonomous Driving, Medical Imaging.

Education

2025 – Present *M.Sc. in Electrical Engineering and Information Technology* — **ETH Zürich**
Major: Signal Processing and Machine Learning. Expected graduation: Sep 2028
Relevant courses: 3D Vision, Deep Learning, Probabilistic Artificial Intelligence

2024 – 2025 *Visiting Student* — **Technical University of Munich (TUM)**
GPA: 10/10. Bachelor's thesis graded **Highest Honors**.

2021 – 2025 *B.Sc. in Artificial Intelligence* — **Universitat Politècnica de Catalunya (UPC)**
Top 10 in the class, GPA: 8.55/10.
1st Class Honors: Voice and Dialog Processing, Advanced Models, HPC, Ethics.

Publications

1. **E. Alegret**, K. Li, S. Wang, S. Liang, M. Niemeyer, S. Gasperini, N. Navab, F. Tombari. *GALA: Guided Attention with Language Alignment for Open-Vocabulary Gaussian Splatting.* International Conference on 3D Vision (**3DV 2026**). arXiv, Project page.

2. S. Wang, K. Li, S. Liang, **E. Alegret**, J. Ma, N. Navab, S. Gasperini. *Visibility-Aware Language Aggregation for Open-Vocabulary Segmentation in 3D Gaussian Splatting.* International Conference on 3D Vision (**3DV 2026**). arXiv, Project page.

Work Experience

Feb 2026 – Present *Student Research Intern* — **ETH Zürich, Computer Vision and Geometry Lab (CVG)**
Advisors: Prof. Federico Tombari, Prof. Marc Pollefeys, Daniel Béla Baráth, Michael Niemeyer and Kunyi Li

- Research on 3D generative models and neural rendering. Co-leading the development of a latent flow-matching model for high-fidelity 3D Gaussian Splatting reconstruction from single-view images, trained at scale on object- and scene-level datasets; targeting **CVPR 2027**.
- Outperforms SOTA models by +1 dB PSNR while preserving geometric accuracy.

Apr 2025 – Jan 2026 *Research Intern* — **TUM, CAMP Lab**
Advisors: Prof. Federico Tombari, Michael Niemeyer, Kunyi Li and Sen Wang

- Co-led the development of GALA (Bachelor's thesis), an open-vocabulary 3D scene understanding framework on 3D Gaussian Splatting that learns instance-level features without supervision and aligns them with language via cross-attention, enabling 2D/3D querying with a low memory footprint.
- Outperformed state-of-the-art methods in open-vocabulary segmentation on real-world benchmarks: +1.4 mIoU on LERF-OVS (2D) and +6–12 mIoU on ScanNet-v2 (3D) over the best prior methods, with 3D queries 300× faster than LangSplat. **Co-first-author paper accepted at 3DV 2026**.

Jul 2024 – Sep 2024 *Research Intern* — **EPFL, LTS4 Lab**
Advisors: Prof. Pascal Frossard and Cédric Vincent-Cuaz

- Built a fine-tuning and evaluation pipeline to benchmark state-of-the-art vision foundation models for cancer detection in X-ray images, and investigated graph-based representations for medical imaging.

Sep 2023 – Jun 2024 — *Research Intern* — **Barcelona Supercomputing Center (BSC)**
Advisor: Prof. Dario García

- Developed Grad-CAM-guided de-biasing for medical-image classifiers and unified multiple medical-imaging datasets into one training pipeline.

Other Projects

Jun 2024 – Jul 2024 — *AI-Driven Clothing Similarity* — **HackUPC**

- Built a deep learning model for visual similarity retrieval using Siamese/Triplet networks with semi-hard negative mining, enabling scalable look-alike and duplicate detection. Code: GitHub.

May 2024 – Jun 2024 — *Explore and Find Node for Robot Exploration* — **UPC**

- Extended the ROS `frontier_exploration` node with YOLO-based object detection, creating a more robust and controlled autonomous exploration system. Code: GitHub.

Skills and Additional Training

Programming	Python, C++, MATLAB
ML & Tools	PyTorch, TensorFlow, Git, Docker, MLflow, ROS, LaTeX
Awards	Academic Excellence Award, Government of Catalonia (2021): top 3% in the university entrance exams
Hackathons	LauzHack (EPFL), hackaTUM (TUM), HackUPC (UPC)
Courses	Convolutional Neural Networks for Visual Recognition (2022), Stanford University; NotOnlyFLOPs: Deep Learning Performance Optimization (2022) and Introduction to Docker Essentials (2024), BSC
Languages	English (C1), Spanish (native), Catalan (native), French (B2), German (beginner)